Andrew Q. Nguyen

US Citizen | (858) 610-4281 | aqnguyen96@gmail.com| LinkedIn | GitHub | Publications

SUMMARY/OBJECTIVE

First generation graduate CS student with network engineering experience with expertise in applied machine learning and cloud infrastructure. Skilled in model management (inference, deployment, evaluation), network analysis with L2/L3 protocol diagnostics (TCP/IP, DHCP, VLANs), and automation (Python, Bash, YAML). I am passionate about optimizing infrastructure and backend systems.

EDUCATION

Northeastern University (GPA: 4.0) [Khoury Align Scholarship Awarded] Seattle, Washington
M.S. *in Computer Science – (DSA, Distributed Systems, Statistics, OOP, ML, Operating Systems).* *Expected Graduation, Dec 2026*

University of California San Diego San Diego, California
B.S. *in Biochemistry and Cell Biology* *Graduated, March 2020*

SKILLS

Programming Languages /Scripting: Python (proficient), Java (proficient), C++ (familiar), YAML, Bash, KQL, RESTful APIs, TDD, Unit Test
Networking, Data Science & ML: OSI Model (L1-L3), TCP/IP, VLANs, DHCP, OSPF/BGP (familiar), ACLs, configuration management, PyTorch, MLflow, Spark, Clustering, Large Language Models (LLM), Inference, Deployment, Evaluation, Databricks, AWS Lambda
DevOps / Databases: Git, Docker, Kubernetes, Jenkins, CI/CD, VMs, KQL DB, Delta Lake, MongoDB, Postgres, Redis, Agile

WORK EXPERIENCES

Centene Corporation Seattle, Washington (Remote)
Machine Learning Engineer Intern *September 2025 – Present*

- **Engineered** a multi-model **evaluation framework** using **MLflow** 3.4 and custom **LLM judges** to benchmark 5 endpoints across **100+** test cases, reducing model selection time by **70%** through automated quality and cost analysis in Unity Catalog.
- **Configured** AI Gateway guardrails on **Databricks** serving endpoints with PII detection and safety filtering, achieving **93%** guardrail effectiveness and **sub-$0.02** cost per request while logging inference data to **Delta Lake** for compliance monitoring.
- **Automated** model **inference** pipelines using MLflow **Pyfunc** eliminating **60%** of cold-start failures for GPU-hosted models

Microsoft Redmond, Washington
Software Engineer Intern *Jun 2025 – Aug 2025*

- **Engineered** a system to automate **configuration management**, ensuring network device compliance and achieving a **25%** reduction in incident response time for configuration drift issues within an **agile** environment using a **TDD** approach.
- **Developed** an automated **YAML workflow** that gathers metadata using **KQL** from a **KQL database,** generate CR/ICM reports via **API** access, then initiates an internal tool to push configuration changes to devices effectively eliminating **90%** of manual work.
- **Maintained** networking systems by **conducting** Tier-2 diagnostics across L2/L3 protocols including **TCP/IP**, **DHCP**, and **ACLs**.

Dr. Alex Yao, San Diego State University San Diego, California
Cloud Data Intern *Dec 2023 – June 2024*

- **Optimized** model accuracy by **12%** by focusing on feature selection and hyperparameter tuning in a product recommendation system using **PyTorch** and e-commerce data stored in **MongoDB**.
- **Enhanced** system scalability and performance by deploying machine learning models via **AWS Lambda** and **Kubernetes**, integrating with **Postgres** and leveraging **Redis** for in-memory caching to optimize data retrieval.
- **Streamlined** model deployment by setting up **CI/CD** pipelines with **Jenkins** and **Docker**, ensuring scalable releases w/minimal errors.

SELECTED PROJECTS

ConquestFour - Qualcomm & Microsoft On-Device AI Hackathon [Demo] Seattle, Washington
Team of 5 Co-Lead Developer *March 2025*

- **Won Second Place out of 28 Teams** creating a **local LLM**-powered Connect Four game using **Mistral-7B (4-bit quantized**).
- **Implemented** Minimax algorithm with Alpha-Beta pruning and **Z3 state validation**, integrated with speech-to-text capabilities using **OpenAI Whisper** Increasing overall player-AI interaction by **75%**.
- **Optimized** performance with **NPU-accelerated** animation reducing game overall processing delay by **60%**.

Semantic Sounds – A Personalized Recommender [Demo] Seattle, Washington
Team of 3 Lead Developer *Dec 2024*

- **Designed** a semantic meaning music recommender system with improved relevancy and **HDBSCAN clustering** effectiveness (Silhouette score: **0.7464**) from base recommender using SHAP-selected features and **sBERT** embeddings.
- **Preprocessed** 60000 Spotify entries achieving regression models' accuracies of **>.5** to identify features influencing song popularity.

CERTIFICATIONS AND ACTIVITIES

Certifications Online
Google Cyber Security Certificate (Linux, MySQL, and Python hands-on labs) *Completed Nov 2024*
Azure Fundamentals (AZ-900) *Completed June 2025*

Selected for Graduate Leadership Institute at Northeastern (GLI) | *Leadership Development* | *Dec 2024* Seattle, Washington

- **Developed** leadership skills through workshops and feedback sessions, resulting in 40% improvement in team performance.